



Semantic search

I have been hearing for some time now that we will soon be seeing many new search engines, and that Google's days are numbered. Given what I have seen and read so far, the developers at Google need not lose much sleep.

I met up recently with Anand Rajaraman, one of the founders of kosmix.com, a new search engine. I have to say that I am deeply unimpressed by the engine — it seems to give very superficial results and leans far too heavily on the sometimes unreliable Wikipedia. But, the developers have some good ideas. The engine is editorialised: it makes use of about 1,00,000 sites that the team has decided are reliable.

This means that if the subject of interest is something they have not considered, or might be found on a site with which they are not familiar, your search will miss out. Now, the developers claim that if you want specific information then perhaps you should use Google, but if you want “content based on the underlying concept” then you should use Kosmix. I have not figured out what that is supposed to mean, but if the subject is not covered in those 1,00,000 sites, then you will get nothing. If the engine cannot find what you want, it serves up Google results — it is very easy to enter a query that produces such a result. This site brings new meaning to the term Limited Area Search Engine. (Look it up; but don't use Kosmix. Actually do, and you'll get only Google results; at least I did while writing this!)

One point on which I agreed with Kosmix's Anand Rajaraman was with regard to semantic search and the semantic web. He is sceptical of them both, and Kosmix is certainly not a semantic search engine. The idea behind such a search engine is that it will appear to understand questions and produce appropriate results. For example, you may be interested in a controversy related to GM foods, and want to ask the question “Why do some criticise the use of a promoter gene from the tobacco mosaic virus?” Type some of those words into Google and you will get a mix of results; go through a few of them and you will almost certainly find the answer. A semantic search engine should, however, get you straight to the right answer.

So, why the scepticism? In the long run, I think we will have semantic interfaces for our computers, the internet, probably our cars and TVs as well. It is just that the methods currently described — most famously by the web founder Tim Berners-Lee — seem to rely too much on people building semantic expression into web content. That might work well for a scientific paper database,

or genome project, but for the general web? Some would have us all entering semantic tags into content so that search engines can glean a deeper understanding of the content and answer more detailed questions, but I just do not see that happening. The work involved in creating the content is just too specialised and too time consuming — and what about all the existing content out there? Who is going to go and convert is all into some kind of semantic XML?

Their soon to be released (May) new system is not a search engine but a “computational knowledge engine”. We will look into this in more detail in a future edition of Digit, but for now, the interesting point is that this engine will accept natural language questions about factual subjects and then compute an answer. Note the verb — compute. It does not look up pre-prepared answers in some kind of database; instead, it computes the answer on the basis of a huge quantity of prepared data.

The limitation here lies in the data that have been chosen and prepared and the algorithms that have been engineered for answering particular types of questions. It is said to be particularly good at scientific subjects and mathematics, but also subjects such as economics and cookery.

Not many details have been released yet, and it is unclear how broad will be the spread of data that Wolfram Alpha will cover; how and how quickly will it incorporate new data; how will it handle facts that are in dispute (something that Wikipedia is spectacularly bad at doing)?

However, the approach that is being taken is very promising, and it looks as though this is going to be a very useful tool. But one that will be used alongside Google, and not instead of it. Search is not the only thing that we need to do with the vast amount of information available on the internet. We do not really need better or different search engines like Kosmix, but we do need other ways of interacting with all that information, and the new Wolfram system looks promising.



Edward Henning, Editor-in-Chief

“Why do some criticise the use of a promoter gene from the tobacco mosaic virus?”

edward.henning@9dot9.in